

# On-Policy Distillation

A practical taxonomy of six research directions

1 Student rollout

2 Teacher feedback

3 Policy update

## 01 OPTIMIZATION

### Training Stabilization

Loss design, clipping, and masking

★ **REOPOLD**      ★ **EOPD**  
★ **Many Faces**      DRKL  
Revisiting OPD      AOPD  
SCOPE      OGLS-SD  
CREDIT      G-OPD  
Sparse-to-Dense

## 02 AUTONOMY

### Teacher-Free Distillation

Self-supervision and iterative improvement

★ **SDFT**      ★ **OPSD**  
Self-Verified      UniSD  
MAIGO      ES-SSD  
ROSD

## 03 INFORMATION

### Privileged Context

The teacher observes richer context

★  $\pi$ -Distill      ★ **SDPO**  
★ **Prefix-OPD**      ★ **MedicalAudit**  
CPCD      HDPO  
OPHSD      Search-E1  
CoDistill      BRTS

## 04 EVALUATION

### Black-Box & Rubric

Reward logic replaces direct logits

★ **ROPD**  
OVD  
MAD-OPD

## 05 EFFICIENCY

### Self-Compression

Smaller token and teacher budgets

★ **CRISP**  
★ **Prefix-OPD** †  
Less-is-More      Lightning OPD  
Sparse-to-Dense †

## 06 TRANSFER

### Cross-Modal & Domain

Vision, video, agents, and applications

★ **Video-OPD**      Vision-OPD  
PRISM      OPSDL  
COPSD      OPSA  
VLA-OPD

★ highlighted work

† cross-family method